Traffic Rule Summary

Sebasitan Maierhofer

January 13, 2020

1 Introduction

This document serves as an overview about formalized traffic rules based on the German Road Traffic Regulation (Straßenverkehrs-Ordnung), in the following abbreviated with StVO, and the Vienna Convention on Road Traffic [1], in the following abbreviated with VCoRT.

The traffic rules formalization process is separated in four steps:

1. **Extracting rules from legal sources**: The rules of the road are specified in the traffic laws of a country or region. However, often traffic rules are not precise and concrete. Therefore, judical decisions which interpret the traffic laws have to be considered. Additionally, it is often the case that several laws have to be considered at the same time because the rules interrelate. Please note that no official translated version of the StVO exist. The translation used in this document is based on the *German Law Archive* website[1].
2. **Concretizing traffic rules and legal judical decisions**: The considered traffic rules and judical decisions have to be combined to a sentence which on the one side describes the situation where the rule is applicable and on the other side concretizes the different legal sources. It can also be the case that a single traffic rule addresses different behaviors and therefore have to be divided (e.g., StVO §4 (1)).
3. **Extracting predicates**: For the logical formalization of traffic rules one needs predicates for the evaluation of certain properties, e.g., allowed speed limit or velocity of preceeding vehicles.
4. **Defining temporal logic formulas**: The last step is to formalize the concretized sentence using the extracted predicates in temporal logic.

We formalize the traffic rules in Linear Temporal Logic (LTL), Metric Temporal Logic (MTL), and Signal Temporal Logic (STL).

The remainder of this paper is structured as follows. Sec. 2 introduces mathematical foundations and different temporal logics. Sec. 4 presents the formalization of traffic rules. The traffic rules are clustered in general traffic rules (Sec. 4.2) applicable in all possible traffic situations, highway traffic rules (Sec. 4.3), urban traffic rules (Sec. 4.4), and cross-country road rules (Sec. 4.5). In Sec. 3 predicates for the temporal logic formulas are defined.

[1]https://germanlawarchive.iuscomp.org/?p=1290

2 Preliminaries

2.1 Road Network

We describe our road network with lanelets, which are atomic, drivable road segments, geometrically represented by a left and right bound, where the bounds are represented by polylines [2]. The set of all lanelets is denoted by $\mathbb{L}$. In the following we refer to a single lanelet as l, where $l \in \mathbb{L}$. The functions $\mathrm{l_{lb}}(l)$, $\mathrm{l_{rb}}(l)$, $\mathrm{l_{begin}}(l)$, and $\mathrm{l_{end}}(l)$ return the left bound, right bound, left and right beginning points, and left and right ending points of a lanelet. The functions

$$\mathrm{adj_l}(l) = \begin{cases} l' & \text{if } \exists\, l' \in \mathbb{L} : \mathrm{l_{lb}}(l') = \mathrm{l_{rb}}(l) \\ \emptyset & \text{otherwise} \end{cases} \tag{1}$$

$$\mathrm{adj_r}(l) = \begin{cases} l' & \text{if } \exists\, l' \in \mathbb{L} : \mathrm{l_{rb}}(l) = \mathrm{l_{lb}}(l') \\ \emptyset & \text{otherwise} \end{cases} \tag{2}$$

$$\mathrm{succ}(l) = \begin{cases} l' & \text{if } \exists\, l' \in \mathbb{L} : \mathrm{l_{end}}(l) = \mathrm{l_{begin}}(l') \\ \emptyset & \text{otherwise} \end{cases} \tag{3}$$

$$\mathrm{pre}(l) = \begin{cases} l' & \text{if } \exists\, l' \in \mathbb{L} : \mathrm{l_{begin}}(l) = \mathrm{l_{end}}(l') \\ \emptyset & \text{otherwise} \end{cases} \tag{4}$$

return the adjacent left lanelet, adjacent right lanelet, successor lanelets, and predecessor lanelets of l, respectively. The curvature of a lanelet at a given path length s_l is given by $\mathrm{curv}(s_\mathrm{l})$.The function $\texttt{speed\_limit}(l)$ returns the maximum speed which a vehicle is allowed to drive on a lanelet. A lanelet can have several attributes which further describe it. The set of all attributes is $\mathbb{A}$. For the highway case, we have $\mathbb{A} = \{main\_carriage\_way, access\_ramp, exit\_ramp, s$
The function $attr(l)$ returns the attributes which are valid for a lanelet. Similar, the left and right bounds of a lanelet can have lanelet markings, with the set of lanelet markings $\mathbb{LM} = \{solid, dashed, broad\_solid, broad\_dashed\}$. The functions $\mathrm{mark}_l(l)$ and $\mathrm{mark}_r(l)$ return the left and right lanelet marking, respectively. We define the lane part in front of a lanelet, in the rear of a lanelet as follows:

$$\mathrm{lane_{front}}(l) = \left\{ l' \,\middle|\, \forall l'', l''' \in (l' \in (\mathrm{succ}(l) \cup \mathrm{lane_{front}}(\mathrm{succ}(l)))) : \mathrm{attr}(l'') = \mathrm{attr}(l''') \right\}, \tag{5}$$

$$\mathrm{lane_{rear}}(l) = \left\{ l' \,\middle|\, \forall l'', l''' \in (l' \in (\mathrm{pre}(l) \cup \mathrm{lane_{rear}}(\mathrm{pre}(l)))) : \mathrm{attr}(l'') = \mathrm{attr}(l''') \right\}. \tag{6}$$

We define the lane to which a lanelet belongs to, using the lanelets in front and rear of another lanelet:

$$\begin{aligned} \mathrm{lane}(l) = \big\{ l' \in (l \ \cup\ \mathrm{lane_{rear}}(l) \ \cup\ \mathrm{lane_{front}}(l)) \,\big|\, \forall l'', l''' \in l' : \mathrm{attr}(l'') = \mathrm{attr}(l''') \\ \wedge\, \mathrm{adj_l}(l') \in \mathrm{succ}((\mathrm{adj_l}(\mathrm{pre}(l'))) \wedge \mathrm{adj_r}(l') \in \mathrm{succ}((\mathrm{adj_r}(\mathrm{pre}(l')))\big\}. \end{aligned} \tag{7}$$

The left and right lane are defined as follows:

$$\mathrm{lane_{left}}(l) = \left\{ l' \,\middle|\, \exists l'' \in \mathrm{lane}(l) : l' = \mathrm{adj_l}(l'') \right\}, \tag{8}$$

$$\mathrm{lane_{right}}(l) = \left\{ l' \,\middle|\, \exists l'' \in \mathrm{lane}(l) : l' = \mathrm{adj_r}(l'') \right\}. \tag{9}$$

The functions

$$\mathrm{lanes_{left}}(l) = \left\{ l' \,\middle|\, \exists l'' \in \mathrm{lane}(l) : l' = \mathrm{adj_l}(l'') \vee \exists l''' \in \mathrm{lanes_{left}}(l'') : l' = \mathrm{adj_l}(l''') \right\}, \tag{10}$$

$$\mathrm{lanes_{right}}(l) = \left\{ l' \,\middle|\, \exists l'' \in \mathrm{lane}(l) : l' = \mathrm{adj_r}(l'') \vee \exists l''' \in \mathrm{lanes_{right}}(l'') : l' = \mathrm{adj_r}(l''') \right\} \tag{11}$$

return all left and right lanes of a given lanelet.

2.2 Vehicle

A vehicle V_i, $i \in \mathbb{N}^+$, consists of a longitudinal, lateral, and signal state $x_{\mathrm{lon}} \in \mathbb{R}^n$, $x_{\mathrm{lat}} \in \mathbb{R}^n$, and $x_{\mathrm{sig}} \subseteq \mathbb{SIG}$, respectively, where $\mathbb{SIG} = \{horn, indicator\_left, indicator\_right, braking\_lights, hazard\_warning\_lights,$ $flashing\_blue\_lights\}$ represents possible active signals of a vehicle. The set of all vehicles is denoted by $\mathbb{V}$.The longitudinal state $x_{\mathrm{lon}} = \begin{bmatrix} s & v & a & j \end{bmatrix}^T$ consists of position s, velocity v, acceleration a, and jerk j. The lateral state $x_{\mathrm{lat}} = \begin{bmatrix} d & \theta & \kappa & \dot{\kappa} \end{bmatrix}^T$ consists of the lateral distance to the reference path Γ, orientation θ, curvature κ, and change of curvature $\dot{\kappa}$. The longitudinal input is $u(t) = \ddot{a}(t)$ and the lateral input is $u(t) = \ddot{\kappa}(t)$. The signal state contains only the subset of currently active elements of $\mathbb{SIG}$. The occupancy of a vehicle for given states x_{lon} and x_{lat} is denoted by $\mathcal{O}(x_{\mathrm{lon}}(t), x_{\mathrm{lat}}(t))$. The function $\mathrm{proj}()$ projects a vehicle to a position s. The functions $\mathrm{front}()$ and $\mathrm{rear}()$ return the front and rear position of a vehicle. All variables associated to the ego vehicle are denoted by the subscript $\square_{\mathrm{ego}}$ and all variables associated to other vehicles are denote by the subscript $\square_{\mathrm{o}}$. For a given initial state, an input trajectory $u(\cdot)$, we introduce the solution of the model $\dot{x} = f(x, u)$ of the ego vehicle over time t as $\xi(t; x_0, u(\cdot))$. We denote a braking input by $u_{brake}(\cdot)$. The time at which a safe state is reached for $u(\cdot)$, given the system dynamics and x_0, is denoted by $t_s(u(\cdot))$; we consider a state to be safe when one can stay in it indefinitely without causing a collision [3].

2.3 Linear Temporal Logic

2.4 Metric Temporal Logic

2.5 Signal Temporal Logic

3 Predicates

3.1 Position

$$\text{occupied\_lanelets}(\mathbb{L}_\text{i}, V_i) = \left\{l' \,\middle|\, l' \in \mathbb{L}_\text{i} : l' \cap \text{proj}(V_i) \neq \emptyset\right\} \tag{12}$$

$$\text{is\_on\_same\_lane\_as\_ego\_vehicle} \iff \text{occupied\_lanelets}(\mathbb{L}_\text{i}, V_\text{ego}) \cap \text{occupied\_lanelets}(\mathbb{L}_\text{i}, V_\text{o}) \neq \emptyset \tag{13}$$

$$\text{is\_behind\_ego\_vehicle} \iff \text{front}(V_\text{o}) \leq \text{rear}(V_\text{ego}) \tag{14}$$

$$\text{is\_in\_front\_of\_ego\_vehicle} \iff \text{front}(V_\text{ego}) \leq \text{rear}(V_\text{o}) \tag{15}$$

$$\text{is\_in\_front\_of\_ego\_vehicle} \iff \text{front}(V_\text{ego}) \leq \text{rear}(V_\text{o}) \tag{16}$$

$$\begin{aligned}\text{vehicles\_in\_front\_of\_ego\_vehicle} = \big\{V' \,\big|\, V' \in \mathbb{V} : \text{occupied\_lanelets}(\mathbb{L}, V')\\ \subseteq \text{lane}_\text{front}(\text{occupied\_lanelets}(\mathbb{L}, V_\text{ego})) \cup \text{occupied\_lanelets}(\mathbb{L}, V_\text{ego})\big\}\end{aligned} \tag{17}$$

3.2 Velocity

Speed limit provided by traffic regulations for a vehicle V_i:

$$v_\text{sl} := \sup\left(\left\{v \,\middle|\, \forall l \in \text{occupied\_lanelets}(\mathbb{L}, V_i) : v = \text{speed\_limit}(l)\right\}\right) \tag{18}$$

The ego vehicle must always be able to stop within the field of view. Therefore, the maximum allowed velocity based on the field of view v_{sr} is chosen such that

$$v_\text{fov} := \begin{aligned}&\text{proj}(\xi(t_s(\tilde{u}_{brake}(\cdot)); x_\text{lon}^\text{sr}, \tilde{u}_{brake}(\cdot))) \leq fov,\\ &x_\text{lon}^\text{sr} = [0, v_\text{fov}, \overline{a}, \overline{j}]^T.\end{aligned} \tag{19}$$

The ego vehicle must always be able reduce its velocity to a upcoming speed limit such that it does not brake abruptly. Therefore, the maximum allowed velocity must be limited such that the braking trajectory does not exceed acceleration and jerk thresholds.

$$\begin{aligned}&v_\text{br} := \sup\left(\left\{v \,\middle|\, x_\text{f} = \xi(t^*; x_0, u(\cdot))) \wedge \forall t \in [t_0, t^*] : a_\text{ego}(t) \geq a^* \wedge j_\text{ego}(t) \geq j^*\right\}\right)\\ &x_0 = [0, v_\text{br}, a_\text{max,ego}, j_\text{max,ego}]^T.\\ &x_\text{f} = [fov, v_\text{sl}^*, 0, 0]^T,\end{aligned} \tag{20}$$

where v_sl^* is the maximum speed limit which is allowed after free driving, e.g., in Germany if there is now active speed limit, the next speed limit is maximal 100 km/h. The velocity limits v_fov and v_br can be calculated offline using binary search.

$$\text{keeps\_lane\_speed\_limit} \iff v_{\text{ego}} \leq v_{\text{sl}} \tag{21}$$

$$\text{keeps\_fov\_speed\_limit} \iff v_{\text{ego}} \leq v_{\text{fov}} \tag{22}$$

$$\text{keeps\_braking\_speed\_limit} \iff v_{\text{ego}} \leq v_{\text{br}} \tag{23}$$

3.3 Safe distance

$$d_{\text{safe}}(v_{\text{rear}}, v_{\text{lead}}) = \begin{cases} d_{\text{safe},1}(v_{\text{rear}}, v_{\text{lead}}), & (a_{\text{min,lead}} > a_{\text{min,rear}}) \wedge (v^*_{\text{lead}} < v_{\text{rear}}) \wedge \\ & (t_{\text{d}} \leq \frac{v_{\text{lead}}}{-a_{\text{min,lead}}}) \wedge (\frac{v_{\text{rear}}}{a_{\text{min,rear}}} > \frac{v^*_{\text{lead}}}{a_{\text{min,lead}}}), \\ d_{\text{safe},2}(v_{\text{rear}}, v_{\text{lead}}), & \text{otherwise}, \end{cases}$$

where

$$v^*_{\text{lead}} = \begin{cases} v_{\text{lead}} - a_{\text{min,lead}} t_{\text{d}}, & t_{\text{d}} \leq \frac{v_{\text{lead}}(t_k)}{-a_{\text{min,lead}}}, \\ 0, & \text{otherwise}, \end{cases}$$

$$d_{\text{safe},1}(v_{\text{rear}}, v_{\text{lead}}) := \frac{(v_{\text{lead}} - a_{\text{min,lead}} t_{\text{d}} - v^2_{\text{rear}})^2}{2(a_{\text{min,lead}} - a_{\text{min,rear}})} + v_{\text{rear}} t_{\text{d}} - v_{\text{lead}} t_{\text{d}} - \frac{1}{2} a_{\text{min,lead}} t^2_{\text{d}},$$

$$d_{\text{safe},2}(v_{\text{rear}}, v_{\text{lead}}) := \frac{v^2_{\text{lead}}}{2a_{\text{min,lead}}} - \frac{v^2_{\text{rear}}}{2a_{\text{min,rear}}} + v_{\text{rear}} t_{\text{d}},$$

and t_{d} denotes the reaction time of the ego vehicle.

$$\text{keeps\_safe\_distance\_front} \iff d_{\text{safe}}(v_{\text{ego}}, v_{\text{o}}) \leq (\text{rear}(V_{\text{o}}) - \text{front}(V_{\text{ego}})) \tag{24}$$

$$\text{keeps\_safe\_distance\_rear} \iff d_{\text{safe}}(v_{\text{o}}, v_{\text{ego}}) \leq (\text{rear}(V_{\text{ego}}) - \text{front}(V_{\text{o}})) \tag{25}$$

3.4 Overtaking

3.5 Unnecessary braking

$$\begin{aligned} &\texttt{unnecessary\_braking} \iff \\ &a_{\text{ego}} < 0 \wedge j_{\text{ego}} < j_{\text{abrupt}} \wedge \nexists V' \in \mathbb{V} : \text{vehicles\_in\_front\_of\_ego\_vehicle} = V' \\ &\vee \exists V' \in \mathbb{V} : \text{vehicles\_in\_front\_of\_ego\_vehicle} = V' \wedge \Delta a < \Delta a_{\text{abrupt}} \wedge \Delta j < \Delta j_{\text{abrupt}} \\ &\vee (v_{\text{sl}}(V') \neq \emptyset \wedge a_{\text{ego}} < a_{\text{abrupt}} \wedge j_{\text{ego}} < j_{\text{abrupt}}) \end{aligned} \tag{26}$$

change acceleration jerk for second case

4 Traffic Rule Formalization

4.1 Traffic Law Terminology

- access ramp / acceleration lane
- off ramp / deceleration lane
- main carriage way
- motor road / motor street / highway
- ego vehicle
- other vehicle

4.2 General Traffic Rules

4.2.1 Distance

Safe distance front	
StVO §4 (1)	Der Abstand zu einem vorausfahrenden Fahrzeug muss in der Regel so groß sein, dass auch dann hinter diesem gehalten werden kann, wenn es plötzlich gebremst wird.
StVO English	A person operating a vehicle moving behind another vehicle must, as a rule, keep a sufficient distance from that other vehicle so as to be able to pull up safely even if it suddenly slows down or stops.
VCoRT §13 (5)	The driver of a vehicle moving behind another vehicle shall keep at a sufficient distance from that other vehicle to avoid collision if the vehicle in front should suddenly slow down or stop.
Concretization	A vehicle moving behind other vehicles within a lane must keep a distance which ensures collision freedom to each of these other vehicles even if one or several of them are suddenly slow down or stop.
Predicates	$\mathrm{keeps\_safe\_distance\_front}$
Comments	Evaluated for all vehicles for which the following condition is valid: $\mathrm{is\_on\_same\_lane\_as\_ego\_vehicle} \land \mathrm{is\_in\_front\_of\_ego\_vehicle}$
LTL	$\mathrm{G}(\mathrm{keeps\_safe\_distance\_front})$
MTL	$\mathrm{G}(\mathrm{keeps\_safe\_distance\_front})$
STL	$\mathrm{G}(d_{\text{safe}}(v_{\text{ego}}, v_{\text{o}}) \leq (\mathrm{rear}(V_{\text{o}}) - \mathrm{front}(V_{\text{ego}})))$

4.2.2 Braking

Unnecessary braking	
StVO §4 (1)	Wer vorausfährt, darf nicht ohne zwingenden Grund stark bremsen.
StVO English	The person operating the vehicle in front must not brake suddenly without a compelling reason.
VCoRT §17 (1)	No driver of a vehicle shall brake abruptly unless it is necessary to do so for safety reasons.
Concretization	A vehicle shall never brake abruptly unless it is necessary to do so for safety reasons. A vehicle brakes abruptly if: 1. It brakes with an acceleration which violates an user-defined threshold and there is no other vehicle, obstacle, or speed limit front of the vehicle; 2. There is an obstacle in front of the vehicle and the acceleration difference and jerk difference between the vehicle and the obstacle violates an acceleration and jerk threshold, respectively; 3. There is an speed limit in front of the vehicle and the acceleration and jerk of the vehicle violate an acceleration and jerk threshold, respectively.
Predicates	$\mathrm{unnecessary\_braking}$
Comments	
LTL	$\mathrm{G(unnecessary\_braking)}$
MTL	$\mathrm{G(unnecessary\_braking)}$
STL	

4.2.3 Velocity

Maximum Speed Limit	
StVO §3 (1), §3 (3), §18 (1), §18 (5), §18 (6)	The legal text is omitted for reasons of space.
StVO English	The legal text is omitted for reasons of space.
VCoRT §13 (2)	Domestic legislation shall establish maximum speed limits for all roads. Domestic legislation shall also determine special speed limits applicable to certain categories of vehicles presenting a special danger, in particular by reason of their mass or their load. They may establish similar provisions for certain categories of drivers, in particular for new drivers.
Concretization	A vehicle shall not exceed the allowed speed limit of the lanes it drives on, shall not exceed the speed limit which is necessary to ensure collision freedom with respect to vehicles outside the field of view, and shall not exceed as speed which is necessary to comfortably ensure traffic regulations.
Predicates	$\mathrm{keeps\_lane\_speed\_limit}$, $\mathrm{keeps\_fov\_speed\_limit}$, $\mathrm{keeps\_braking\_speed\_limit}$
Comments	The traffic laws do not provide concrete information about speed limit, e.g., the VCoRT passes the task of defining speed limits to national lawmakers. The concretization assumes that a mapping of speed limits provided by traffic signs to lanes is given. If no speed limit is given, the speed limit caused by the field of view (traffic sign based speed limit set to infinity) and a speed limit which ensures comfortable braking for traffic regulations (e.g., upcoming speed limit) must be considered. The velocity with respect to vehicles inside the field of view is considered in the safe distance traffic rule concretization.
LTL	$\mathrm{G}(\mathrm{keeps\_lane\_speed\_limit} \wedge \mathrm{keeps\_fov\_speed\_limit} \wedge \mathrm{keeps\_braking\_speed\_limit})$
MTL	$\mathrm{G}(\mathrm{keeps\_lane\_speed\_limit} \wedge \mathrm{keeps\_fov\_speed\_limit} \wedge \mathrm{keeps\_braking\_speed\_limit})$
STL	$\mathrm{G}(v_{\mathrm{ego}} \leq v_{\mathrm{sl}} \wedge v_{\mathrm{ego}} \leq v_{\mathrm{fov}})$

Minimum Speed Limit	
StVO §3 (2)	Ohne triftigen Grund dürfen Kraftfahrzeuge nicht so langsam fahren, dass sie den Verkehrsfluss behindern.
StVO English	No motor vehicle must, without good reason, travel so slowly as to impede the flow of traffic.
VCoRT §13 (2)	Similar to maximum speed limit.
Concretization	The relative velocity of a vehicle with respect to leading and following vehicles should not exceed a certain threshold.
Predicates	$\mathrm{keeps\_min\_speed\_limit}$
Comments	
LTL	$\mathrm{G}(\mathrm{keeps\_min\_speed\_limit})$
MTL	$\mathrm{G}(\mathrm{keeps\_min\_speed\_limit})$
STL	$\mathrm{G}(v_{\mathrm{o}}^{\mathrm{min}} - v_{\mathrm{ego}} \leq v_{\mathrm{min}}^{\mathrm{slow}})$

4.3 Highway Traffic Rules

Driving right faster than left	
StVO §7 (2), §7 (2a), §7a	Ist der Verkehr so dicht, dass sich auf den Fahrstreifen für eine Richtung Fahrzeugschlangen gebildet haben, darf rechts schneller als links gefahren werden. Wenn auf der Fahrbahn für eine Richtung eine Fahrzeugschlange auf dem jeweils linken Fahrstreifen steht oder langsam fährt, dürfen Fahrzeuge diese mit geringfügig höherer Geschwindigkeit und mit äußerster Vorsicht rechts überholen. Auf Ausfädelungsstreifen darf nicht schneller gefahren werden als auf den durchgehenden Fahrstreifen. Stockt oder steht der Verkehr auf den durchgehenden Fahrstreifen, darf auf dem Ausfädelungsstreifen mit mäßiger Geschwindigkeit und besonderer Vorsicht überholt werden. Gehen Fahrstreifen, insbesondere auf Autobahnen und Kraftfahrstraßen, von der durchgehenden Fahrbahn ab, darf beim Abbiegen vom Beginn einer breiten Leitlinie (Zeichen 340) rechts von dieser schneller als auf der durchgehenden Fahrbahn gefahren werden. Auf Ausfädelungsstreifen darf nicht schneller gefahren werden als auf den durchgehenden Fahrstreifen. Stockt oder steht der Verkehr auf den durchgehenden Fahrstreifen, darf auf dem Ausfädelungsstreifen mit mäßiger Geschwindigkeit und besonderer Vorsicht überholt werden.
StVO English	If the density of traffic has resulted in queues of vehicles in the lanes of one carriageway, traffic on the right (nearside lane, middle lane) may move faster than traffic on the left (offside lane, middle lane). If, on a carriageway for one direction of traffic, a vehicle queue has formed and come to a standstill or is moving at low speed in the left-hand lane, vehicles may overtake this queue on the right at a slightly higher speed and with the utmost care. In deceleration lanes, vehicles must not travel faster than traffic on the main carriageway. If traffic on the main carriageway is moving slowly or is stationary, vehicles in a deceleration lane may overtake at a moderate speed and with great care. Where lanes diverge from the main carriageway, especially on motorways and motor roads, vehicles turning off may, from the beginning of and to the right of a lane marking with broad lines (sign 340), travel faster than traffic on the main carriageway. On motorways and other roads outside built-up areas, vehicles may travel faster in acceleration lanes than traffic on the main carriageway.

Driving right faster than left	
VCoRT	
Concretization	A vehicle shall not drive faster than any vehicle on its left lanes. A exception is, if 1. the vehicle on the left lane is in a congestion; 2. the vehicle on the left lane is on the main carriage way and the vehicle on the right lane is on a off ramp; 3. the left vehicle and right vehicle are separated by a broad lane marking; 4. the right vehicle is within an access ramp. For 1. and 2., the vehicle on the right lane is allowed to drive faster than the vehicle on the left lane with a slightly higher speed.
Predicates	drives_faster_than_vehicle_left, congestion_left, other_vehicle_left_of_broad_lane_marking, drives_max_with_slightly_higher_speed, right_of_broad_lane_marking
Comments	Assumption: Deceleration lane always right of main carriage way. Case two can be neglected -¿ same as case one
LTL	$\mathrm{G}(\neg\, \text{drives\_faster\_than\_vehicle\_left}$ $\lor(\text{congestion\_left} \land \text{drives\_max\_with\_slightly\_higher\_speed})$ $\lor(\text{right\_of\_broad\_lane\_marking} \land \text{other\_vehicle\_left\_of\_broad\_lane\_marking})$ $\lor(\text{on\_access\_ramp} \land \neg\, \text{congestion\_left}))$
MTL	$\mathrm{G}(\neg\, \text{drives\_faster\_than\_vehicle\_left}$ $\lor(\text{congestion\_left} \land \text{drives\_max\_with\_slightly\_higher\_speed}))$
STL	

Emergency Lane	
StVO §3 (2)	Sobald Fahrzeuge auf Autobahnen sowie auf Außerortsstraßen mit mindestens zwei Fahrstreifen für eine Richtung mit Schrittgeschwindigkeit fahren oder sich die Fahrzeuge im Stillstand befinden, müssen diese Fahrzeuge für die Durchfahrt von Polizei- und Hilfsfahrzeugen zwischen dem äußerst linken und dem unmittelbar rechts daneben liegenden Fahrstreifen für eine Richtung eine freie Gasse bilden.
StVO English	As soon as vehicles on motorways and roads outside a built-up area with at least two lanes for one direction start to move at walking pace or come to a standstill, these vehicles must leave a gap for one direction between the lane on the far left and the lane immediately adjacent to it on the right to allow police and emergency vehicles to pass.
VCoRT	
Concretization	A vehicle in a congestion outside urban roads has to drive leftmost if it is in the leftmost lane and has to drive rightmost otherwise. not exceed a certain threshold.
Predicates	keeps_min_speed_limit
Comments	
LTL	$\mathrm{G}(\text{keeps\_min\_speed\_limit})$
MTL	$\mathrm{G}(\text{keeps\_min\_speed\_limit})$
STL	$\mathrm{G}(v_{\mathrm{o}}^{\mathrm{min}} - v_{\mathrm{ego}} \leq v_{\mathrm{min}}^{\mathrm{slow}})$

4.4 Urban Traffic Rules

4.5 Cross-Country Road Traffic Rules

References

[1] Economic Comission for Europe: Inland Transport Committee, "Vienna Convention on Road Traffic," Nov. 1968.

[2] P. Bender, J. Ziegler, and C. Stiller, "Lanelets: Efficient map representation for autonomous driving," in *Proc. of the IEEE Intelligent Vehicles Symposium*, 2014, pp. 420–425.

[3] C. Pek and M. Althoff, "Efficient Computation of Invariably Safe States for Motion Planning of Self-Driving Vehicles," in *Proc. of the IEEE Int. Conf. on Intelligent Robots and Systems*, 2018, pp. 3523–3530.